Backing Up Data with FOSS Tools

Given that backing up data is an essential part of our digital activities, which tools would be best for the task? Naturally, our choice would be the open source variants! The author gives readers a choice of three such tools.

In a world of rapidly evolving technologies, data is everything. Organisations, and even individuals like you and me, are heavily dependent on data for our day-to-day work. System failures, corrupt hard drives, virus infections, etc, are a few of the causes of data loss. Protecting data from these hazards is not that difficult. An effective way to do so is to maintain a regular backup of data, which will ensure that at any given time you have at least one fail-safe copy of your data that you can rely upon.

The tools discussed below have been chosen for their simplicity and as alternatives to advanced backup solutions.

Data backup terminology

There are certain terms related to data backup software that you should be familiar with, to help you in understanding the tools discussed below.

Synchronisation: Maintaining two or more copies of files and folders (data), at different locations, such that any changes made (to data) in one location are reflected in others.

Mono-directional synchronisation: Synchronisation happens in one direction only, i.e., changes made in Location A will be reflected in Location B but not vice versa.

Bi-directional synchronisation: Synchronisation happens in both directions—changes made in either location will reflect in the other.

Full backup: All files and folders will be copied to the backup location irrespective of whether they are changed or not. In a full backup, files are compressed and can be password protected.

Mirror backup: This is identical to the full backup except that it does a straight copy of the original location without compression or password protection.

Incremental/update backup: Only the files that were created or modified after the last backup will be copied or backed up.

Mirror (incremental) backup: As the name implies, this is a combination of a mirror and an incremental backup. It copies only the newly created or modified files to the backup location without any compression or password protection.

Contribute: Only new files will be copied to the backup location.

Schedule: A time table for which backup job to run. This can be daily, weekly, monthly, annually, etc. Once a schedule is set, the tool will automatically start the backup job at the defined interval.

With that out of the way, let us get down to exploring FOSS data backup tools.

Duplicati

Developed by Kenneth Skovhede, Duplicati is a powerful, easy to use backup client. It is free, open source, feature-packed and can work on multiple platforms. The best thing about this tool is that it provides an option to back up data onto cloud storage.